



# ASME

## FEDSM 2026

FLUIDS ENGINEERING DIVISION  
SUMMER MEETING

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HYATT REGENCY BELLEVUE | BELLEVUE, WA, USA

# RESIDUAL MANIFOLDS AND CLOUD UNCERTAINTY

A UNIFIED FRAMEWORK FOR PHYSICAL SYSTEM PREDICTION  
VALIDATED IN AERODYNAMIC COEFFICIENT PREDICTION

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# Motivation

- CFD and wind tunnel experiments are comparatively accurate but costly.
- Reduced order methods are efficient but miss viscous and separation effects.
- In many engineering applications, only a small number of higher-fidelity observations are available.

**Can we learn the missing physics from sparse data while retaining the efficiency and interpretability of a baseline?**



# Prior Art and Gap

Deep learning surrogates have to learn the complete relationship and can overfit when training data are sparse.

PINN require governing-equation constraints or explicit parameterization of physical perturbations.

Manifold learning exploit low-dimensional structure, but are not typically used as corrections to an inexpensive physics baseline.

**Our method learns the discrepancy from a baseline, lifts the operating variables into a richer feature space, and organizes the residual behavior within a structured latent representation.**



# Main Contributions

- Evaluate residual learning for sparse correction of a fast physics baseline.
- Lift the operating variables with polynomial and radial-basis features before constructing the local latent representation.
- Use a compact ridge-base corrector to remove remaining systematic bias.

Deliver a data-efficient and extensible hybrid prediction framework.



# Framework Overview

Fast physics baseline

Feature lifting with polynomial and radial-basis functions

Quantile binning and local principal-component analysis

Latent residual prediction with an MLP

Ridge solved correction

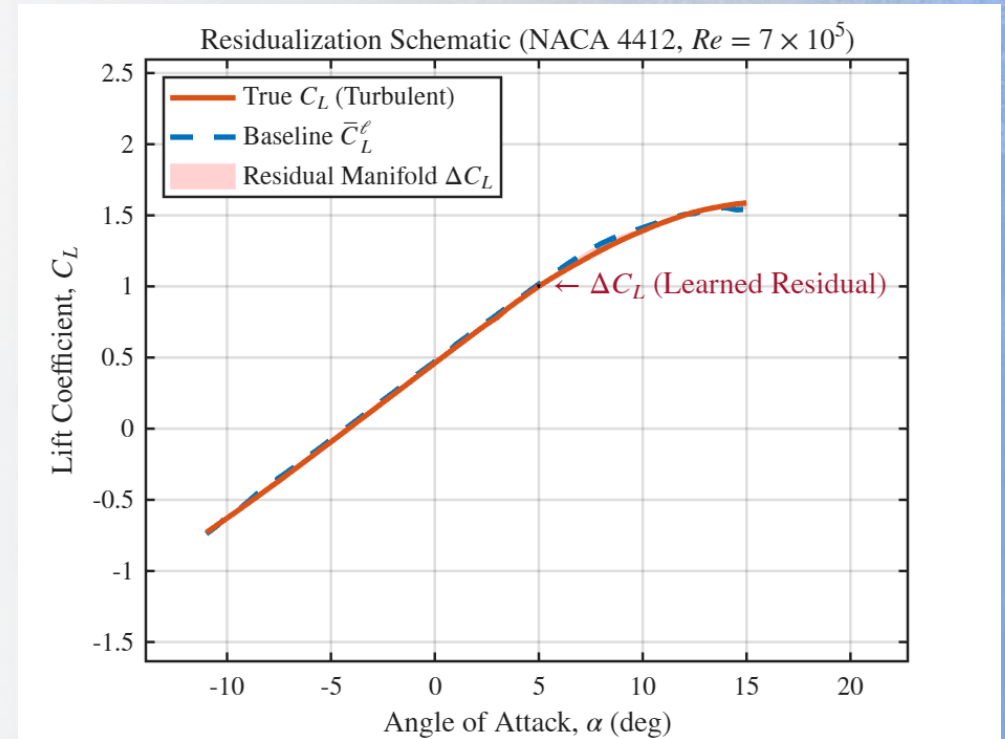
Baseline + learned correction = final coefficients



# Residual Formulation

- A fast viscous-inviscid solver supplies the clean-flow baseline.
- The learner targets only discrepancies from turbulence, viscous interaction, and separation.
- Residual anchoring reduces the burden on the small-data model.

$$\Delta C(\alpha, Re) = C_{true} - C_l$$



Residualization schematic (paper, Fig. 1)



# Latent Decoder and Cloud Correction

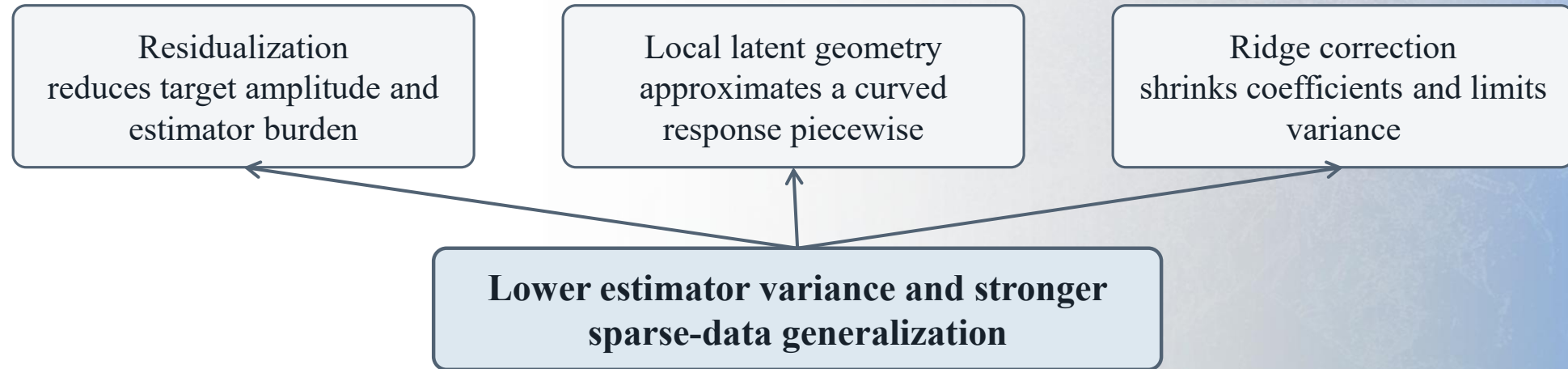
- Feature lifting increases expressive capacity before projection.
- Quantile binning and local PCA form a piecewise-linear approximation to curved residual geometry.
- Validation preserves weak but sensitive latent directions that variance ranking alone may discard.
- The MLP captures local nonlinearity; ridge shrinkage limits coefficient variance and mean bias.

$$\mathbf{Z}(\alpha, R_e) = [\Phi_X(\alpha, R_e) - \mu_\Phi] \mathbf{W}_{\text{proj}}$$

$$\widehat{\Delta \mathbf{C}} = g_{\text{MLP}}(\mathbf{Z}) + \mathbf{W}_c^T \begin{bmatrix} 1 \\ \mathbf{Z} \\ \tanh(\mathbf{Z}) \end{bmatrix}, \quad \widehat{\mathbf{C}} = \overline{\mathbf{C}}_\ell + \widehat{\Delta \mathbf{C}}$$



# Why the hybrid generalizes from sparse data



*The framework replaces one high-variance global fit with controlled, lower-complexity subproblems.*



# Experiment Design

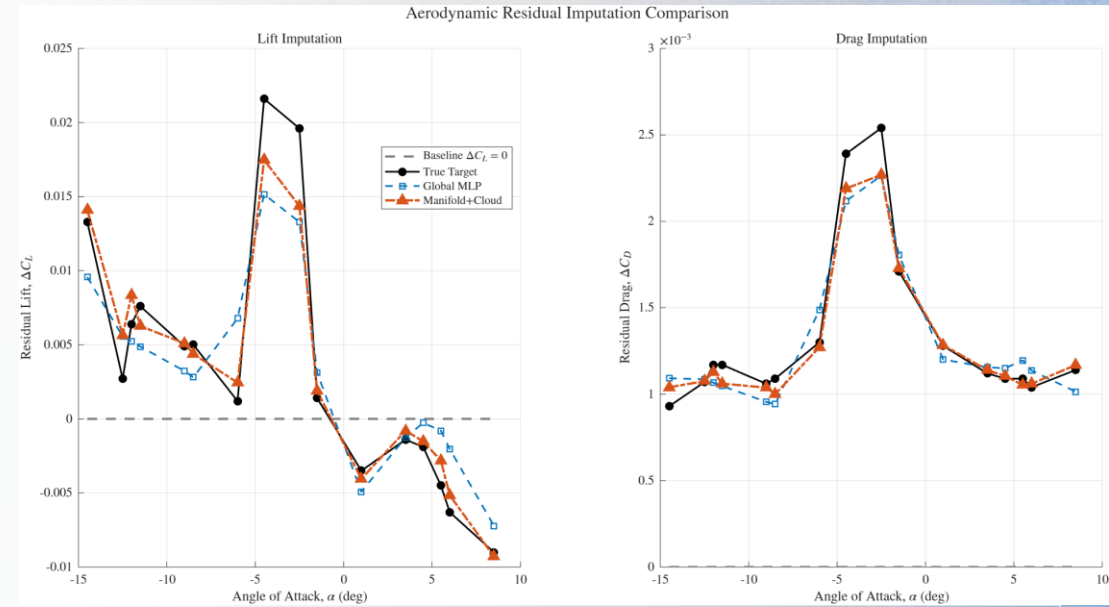
Sparse aerodynamic-matrix imputation

- Three NACA airfoils: 0012, 23014, and 23024
- Envelope:  $-15^{\circ}$  to  $15^{\circ}$ ; Reynolds numbers from 700,000 to 7,000,000
- Clean baseline: critical transition factor 9; turbulent target: 3
- Holdouts: 25, 50, 75, and 100 rows (random)
- 50 trials; transforms and models fitted only on training data



# Results - Accuracy

- Direct global model: 0.722
- Hybrid model: 0.957
- Unexplained variance reduced by 82.9%
- Root-mean-square and mean absolute errors reduced by about 61%
- Hybrid methods consistently outperform direct learning.



Representative sparse polar reconstruction (paper, Fig. 3)



# Hybrid gains are consistent across all three airfoils

| Airfoil    | Global<br>coef. det. | Manifold + MLP<br>coef. det. | MLP + Cloud<br>coef. det. | RMSE<br>reduction | MAE<br>reduction |
|------------|----------------------|------------------------------|---------------------------|-------------------|------------------|
| NACA 23024 | 0.643                | 0.949                        | 0.952                     | 66.1%             | 65.3%            |
| NACA 0012  | 0.866                | 0.965                        | 0.967                     | 49.8%             | 54.7%            |
| NACA 23014 | 0.658                | 0.951                        | 0.953                     | 67.3%             | 67.9%            |

The improvement is not driven by one geometry: both cambered airfoils and the symmetric NACA 0012 case benefit.



# Hybrid accuracy remains high under severe holdout

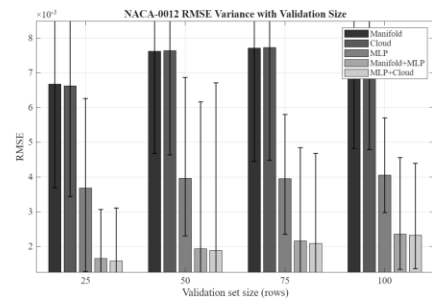
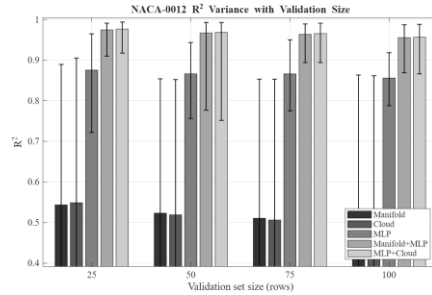
| Model          | Coef. det.<br>25 rows | Coef. det.<br>100 rows | Relative<br>change | RMSE<br>25 rows | RMSE<br>100 rows | RMSE<br>change |
|----------------|-----------------------|------------------------|--------------------|-----------------|------------------|----------------|
| Global MLP     | 0.751                 | 0.685                  | -8.7%              | 4.79E-03        | 5.80E-03         | +21.0%         |
| Manifold + MLP | 0.965                 | 0.939                  | -2.7%              | 1.67E-03        | 2.51E-03         | +50.4%         |
| MLP + Cloud    | 0.968                 | 0.940                  | -2.9%              | 1.60E-03        | 2.47E-03         | +54.4%         |

Even with 100 rows withheld, both structured hybrids retain approximately 0.94 coefficient of determination.

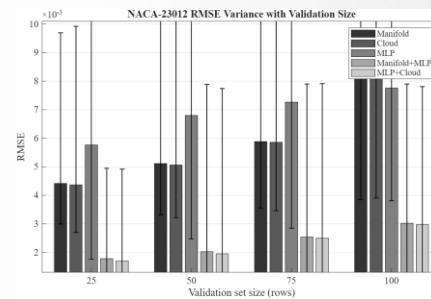
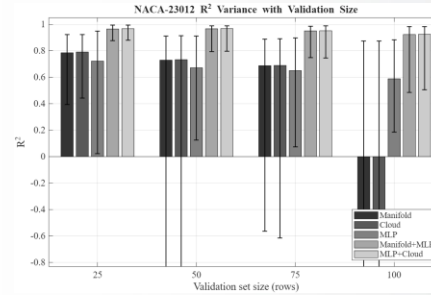


# Appendix trends confirm repeatable performance

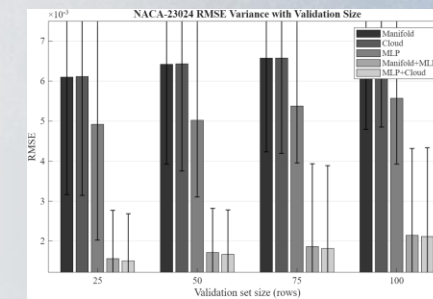
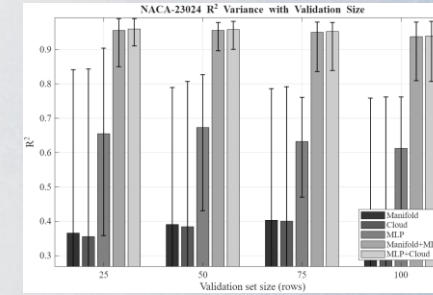
NACA 0012



NACA 23014



NACA 23024



Across validation sizes, the structured models remain consistently stronger than the direct global baseline.



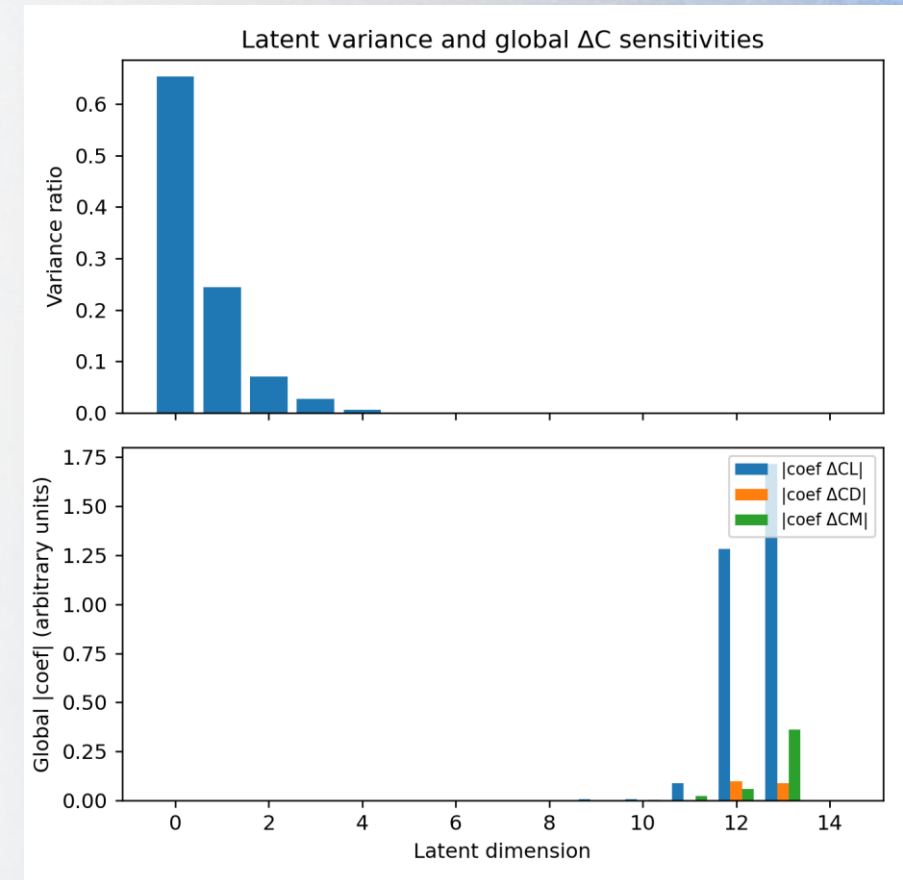
# Results - Robustness and Latent Sensitivity

At 100 withheld rows

- The hybrid retains a coefficient of determination of 0.940.
- The direct global model falls to 0.685.

Latent variance concentrates in leading modes, while residual sensitivity can peak at higher-index coordinates.

Weak latent directions may carry important transition or separation information.



*Latent variance and sensitivity example (paper, Fig. 4)*



# Limitations and Scope

- Validation uses XFOIL-generated clean and turbulent cases rather than wind-tunnel measurements or high-fidelity computational fluid dynamics.
- The demonstration is limited to three two-dimensional NACA airfoils and interpolation within the sampled operating envelope.
- The cloud stage is a deterministic ridge correction; calibrated confidence intervals are not yet produced.
- Hyperparameter selection and repeated trials increase training cost.



# Conclusions and Implications

- Residual learning isolates unresolved physics instead of relearning the full aerodynamic map.
- Lifted features and local principal-component analysis organize sparse observations into a structured latent representation.
- The hybrid model reaches an average coefficient of determination of 0.957 and lowers absolute error by about 61% relative to direct global learning.
- Accuracy remains high under severe holdout, supporting sparse experimental-matrix completion.

Next step: validate with wind-tunnel or high-fidelity data, then extend to three-dimensional wings and multi-fidelity datasets.



# Thank you!

Questions?

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